A Semi-Supervised Multiscale Convolutional Sparse Coding-Guided Deep Interpretable Network for Hyperspectral Image Change Detection

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***Abstract*— Deep learning (DL) has increasingly become the mainstream technology for hyperspectral image change detection (HSI-CD). However, these methods lack transparency and often overlook priors in hyperspectral images (HSIs), making it difficult to extract more generalizable features directly from them. Moreover, the performance of DL-based methods typically depends heavily on a large corpus of high-quality labeled data, which is often impractical and expensive in real-world scenarios, particularly for complex HSIs. To address these issues, we propose a semi-supervised deep interpretable network for HSI-CD. Specifically, by applying structured sparse prior constraints, we propose a multiscale convolutional sparse coding (MSCSC) model to capture shared and private sparse coefficients (SSC and PSCs) across different scales, extracting multiscale features while reducing redundancy. We then unfold the proposed MSCSC model to establish an MSCSC-guided deep interpretable network (MSCSCNet) that serves as the encoder, namely, MSCSCNet, in which each network module is model-driven, enhancing the transparency of internal mechanisms and extracting more fundamental features. In addition, we devise a two-stage semi-supervised training strategy for MSCSCNet using limited labeled data, combining a change-sensitive teacher–student self-distillation (CS-TSSD) paradigm with a novel loss function to reduce the annotation dependency and remain sensitive to change components. The proposed method not only enhances the transparency of multiscale feature extraction but also accomplishes semi-supervised learning to extract meaningful representations, effectively integrating the advantages of model-driven and data-driven approaches. Comparative experiments on benchmark datasets demonstrate the effectiveness of our method over the existing approaches. Code is available at https://github.com/Jiahuiqu/MSCSCNet**

***Index Terms*— Change detection (CD), deep interpretable network, hyperspectral image (HSI), multiscale convolutional sparse coding (MSCSC), semi-supervised learning.**

NOMENCLATURE

ADMM	Alternating direction method of multipliers.
AV	Auxiliary variable.
CD	Change detection.
CS-TSSD	Change-sensitive teacher–student self-distillation.
DL	Deep learning.
DV	Dual variable.
EMA	Exponential moving average.
HSI	Hyperspectral image.
MLP	Multilayer perceptron.
MSCSC	Multiscale convolutional sparse coding.
PSC	Private sparse coefficient.
PSCUM	Private sparse coefficient update module.
PV	Private variable.
SC	Sparse coefficient.
SL	Supervised learning.
SSC	Shared sparse coefficient.
SSCUM	Shared sparse coefficient update module.
SSL	Self-supervised learning.
SV	Shared variable.

I. INTRODUCTION

AS A means of detection and identification, remote sensing technologies collect information from the Earth's surface noncontactly using sensors on aircraft or satellites. Among them, CD technology analyzes images of the same region obtained at different times to identify and quantify surface changes. This technology holds extensive application value in fields, such as urban planning, natural disaster assessment, and environmental protection. In recent years, HSIs have received widespread attention for their ability to capture a broad spectral range [1], [2], offering richer and finer spatial and spectral information, which enables it sensitive to subtle and intricate changes, thus becoming a significant focus in CD research [3].

Early CD research includes image algebra, transformation-based, and classification-based methods. Change vector analysis (CVA) [4], a paradigmatic image algebra method, identifies changes using pixelwise Euclidean distances

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but is noise-sensitive due to neglecting spatial context. Transformation-based methods, such as principal component analysis (PCA) [5], slow feature analysis (SFA) [6], and iteratively reweighted multivariate alteration detection (IR-MAD) [7], are commonly used to reduce data dimensionality or enhance separability. However, these methods may lose useful information during the dimensionality reduction. Classification-based techniques include postclassification and compound classification [8]. Postclassification separately classifies two images, while compound classification directly classifies overlaid bitemporal HSIs, both relying heavily on classifier design and facing cumulative or omissive errors.

Over recent years, DL has become widely adopted in computer vision and remote sensing [9], [10], [11], [12], [13]. In CD, DL-based methods have dominated. For instance, Wang et al. [14] introduced general end-to end 2-D convolutional neural network (GETNET), a 2-D convolutional neural network (CNN) that spectrally unmixes HSIs to produce a mixed affinity matrix for feature extraction. Mou et al. [15] developed recurrent CNN (ReCNN), extracting spectral–spatial features from bitemporal images using CNNs, which are then processed by recurrent neural network (RNN), gated recurrent unit (GRUs), or LSTMs for temporal dependencies. Qu et al. [16] proposed ML-EDAN, a multilevel encoder–decoder attention network with an long short-term memory (LSTM) subnetwork for temporal dependencies. Luo et al. [17] proposed MSDFFN, accentuating subtle changes across scales in bitemporal HSIs and enhancing feature representation. Wang et al. [18] introduced SST-Former, a transformer network for joint spectral–spatial–temporal encoding.

Despite the significant success of DL-based methods, they remain empirical, lack clear mathematical guidance, and often overlook the intrinsic sparse priors of HSIs due to spectral and spatial redundancy [19]. Their "black-box" nature limits interpretability and theoretical advancements [20]. In contrast, model-based approaches offer better explainability and generalization with fewer training data but often learn shallow features. Consequently, combining DL and model-based methods can enhance performance, providing interpretability and better generalization without relying on large datasets. Nevertheless, interpretability-driven network designs in CD lag behind fields, such as denoising [21], super-resolution [22], and fusion [23]. On the other hand, DL techniques often require extensive annotated data, which is impractical for complex HSIs. In CD, several SSL-based studies have been conducted. Hu et al. [24] proposed the HyperNet, selecting unchanged pixels as positive pairs to learn invariant features and separate unchanged from changed areas. However, this focus on invariance neglects crucial change information. In addition, methods in [25] and [26] use image pairs from the same locations (scenes) as positive samples and different locations (scenes) as negative samples, aiming to minimize positive pair distances and maximize negative pair distances in the embedding space. This approach may incorrectly push apart representations of the same land cover at different locations, leading to inaccurate learning. Therefore, it is important to develop new learning strategies that maintain the focus on unchanged information while more effectively identifying and emphasizing change components.

In this article, we present a semi-supervised MSCSC-guided deep interpretable network (MSCSCNet) for HSI change detection (HSI-CD). Our approach unfolds a novel MSCSC model to capture SSC and PSCs across scales, establishing a deep interpretable network. To train MSCSCNet with limited annotated data, we propose a two-stage semi-supervised strategy: first, SL is used to "warm up" the encoder with robust initialization, followed by a second stage that integrates CS-TSSD with SL, allowing the network to learn from abundant unlabeled data while maintaining focus on CD. The key contributions of this article can be summarized as follows.

1) We propose a semi-supervised MSCSCNet architecture for HSI-CD, which can effectively extract multiscale features with clear physical interpretability while learning useful information from a large amount of unlabeled data and enhancing sensitivity to change components.
2) A novel MSCSC model is proposed to capture both SSC and PSCs across various scales, enhancing the interpretability of multiscale feature extraction while reducing information redundancy. When integrated with DL, the model is then unfolded to construct a deep interpretable network, MSCSCNet, which effectively extracts deep and physically interpretable multiscale features.
3) We devise a two-stage semi-supervised training strategy that employs a hybrid training mechanism of SL and SSL. For SSL, we introduce the CS-TSSD method, featuring a novel loss function. This training approach enables the proposed model to learn from a vast amount of unlabeled data while improving its capability of identifying changes and receiving accurate guidance from limited labeled data.

The remainder of this article is structured as follows. Section II details the principles and components of the proposed method. Section III presents comparative experimental results on three benchmark datasets, along with ablation studies of our method. Section IV concludes this article.

II. Methodology

In this section, as depicted schematically in Fig. 1, we present a semi-supervised deep interpretable neural network architecture for HSI-CD. First, we introduce the MSCSC model, which captures SSC and PSCs across different scales, and detail its iterative update algorithm. Next, combining DL techniques, we explain how this algorithm is unfolded into MSCSCNet, serving as the encoder. Finally, we discuss the proposed two-stage semi-supervised training strategy in detail. Sections II-A–II-C will elaborate on these components.

A. Multiscale Convolutional Sparse Coding Model

Due to their high-dimensional complexity, HSIs exhibit significant spectral and spatial correlations, which inherently endow them with sparse priors [19]. Sparse representation, including dictionary learning and sparse coding, effectively addresses these challenges [27], [28]. However, most methods vectorize input images, disrupting spatial structure and losing

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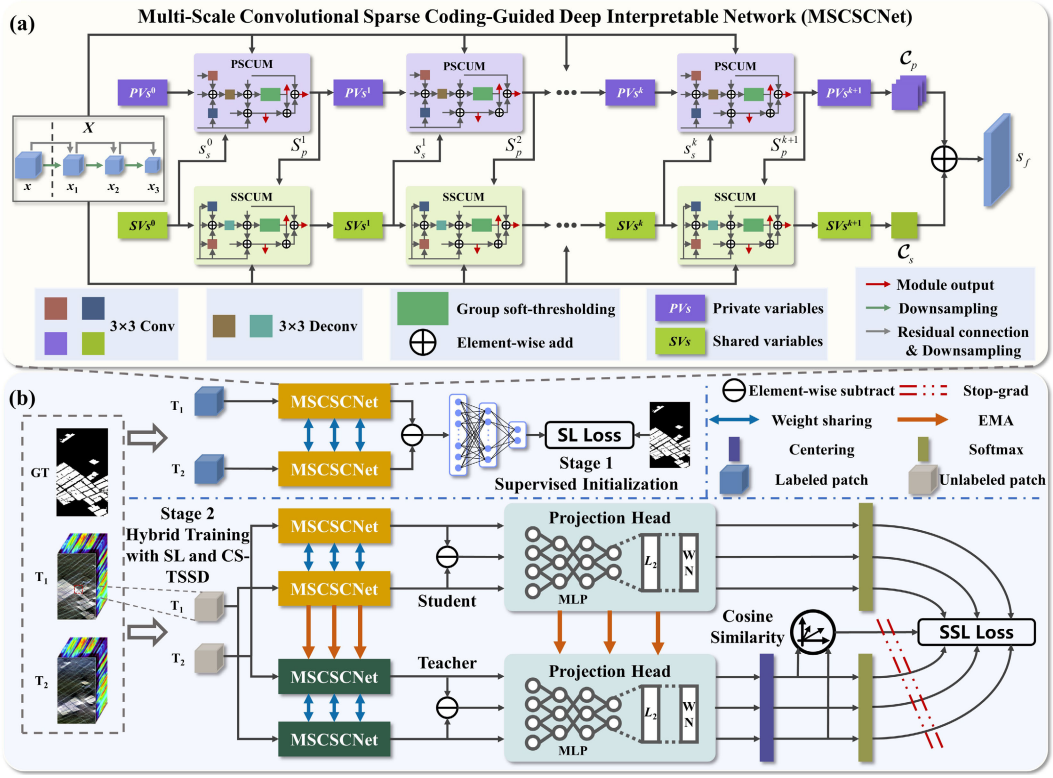


Fig. 1. Overview of the proposed semi-supervised deep interpretable network. (a) Architecture of MSCSCNet. (b) Two-stage semi-supervised training strategy consisting of supervised initialization and hybrid training with SL and CS-TSSD.

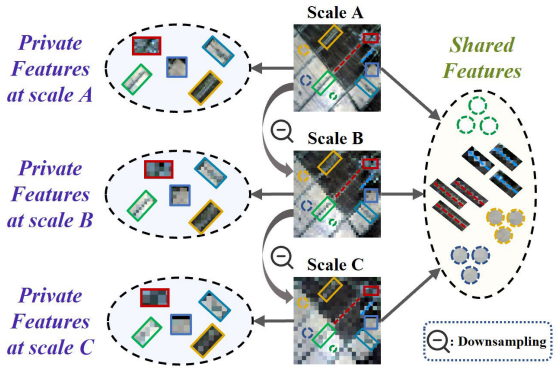


Fig. 2. Illustration of shared and private features across different scales.

critical information, which is particularly detrimental to CD tasks relying on spatial context. Given the varied sizes of changed and unchanged regions within a scene, extracting multiscale features in CD is crucial. To this end, we introduce a novel MSCSC model to capture SSC and PSCs across scales and integrate them into a fused SC.

The image pyramid theory [29] posits that at different scales, an image contains varying levels of detail and structural complexity, with smaller scales corresponding to coarser structures. However, certain key patterns may exist at any scale within the image. Based on this, we consider two cases for multiscale feature extraction. As illustrated in Fig. 2, the same scene at different scales exhibits unique private features, displaying varying textures and structural details. On the other hand, cross-scale shared features exist across different scales, manifested as similar overall contours, consistent basic textures and structures, and scale-invariant directional features. Therefore, different PSCs can be used to represent these private features, while a single, consistent SSC is utilized for the cross-scale shared features. In this way, the MSCSC method enhances feature discrimination by distinguishing and encoding private and shared features, enriching the coefficient space with multiscale information. This reduces feature redundancy and repetitive extraction across scales, resulting in more compact and goal-oriented coefficient features. Furthermore, this separation increases model interpretability, making the decision-making process more transparent.

To model the multiscale feature extraction, feature maps at three different scales are considered for a hyperspectral input patch $\boldsymbol{x} \in \mathbb{R}^{C\times h\times w}$, denoted as $\boldsymbol{x}_1 \in \mathbb{R}^{C\times h_1\times w_1}$, $\boldsymbol{x}_2 \in \mathbb{R}^{C\times h_2\times w_2}$, and $\boldsymbol{x}_3 \in \mathbb{R}^{C\times h_3\times w_3}$. The MSCSC model is formulated as follows:

$$\boldsymbol{x}_1 = \boldsymbol{\mathcal{D}}_s^1 \otimes \boldsymbol{s}_s + \boldsymbol{\mathcal{D}}_p^1 \otimes \boldsymbol{s}_p^1 \tag{1}$$

$$\boldsymbol{x}_2 = \boldsymbol{\mathcal{D}}_s^2 \otimes \boldsymbol{s}_s + \boldsymbol{\mathcal{D}}_p^2 \otimes \boldsymbol{s}_p^2 \tag{2}$$

$$\boldsymbol{x}_3 = \boldsymbol{\mathcal{D}}_s^3 \otimes \boldsymbol{s}_s + \boldsymbol{\mathcal{D}}_p^3 \otimes \boldsymbol{s}_p^3 \tag{3}$$

where $\otimes$ represents the convolutional operation, $\boldsymbol{s}_s \in \mathbb{R}^{P\times h_2\times w_2}$ denotes the SSC, $\boldsymbol{s}_p^i \in \mathbb{R}^{P\times h_i\times w_i}$ denotes the PSCs, $\boldsymbol{\mathcal{D}}_s^i \in \mathbb{R}^{C\times P\times k\times k}$ denotes the shared filters, and $\boldsymbol{\mathcal{D}}_p^i \in \mathbb{R}^{C\times P\times k\times k}$ $(i = 1, 2, 3)$ represents the private filters. In addition, h_i and w_i $(i = 1, 2, 3)$ represent the height and width at different scales, respectively, C is the number of bands, and k is the convolution kernel size. P is the number of channels in the SCs, determining the representational capacity of the model. We strategically set the spatial dimensions of the SSC to align with those of $\boldsymbol{x}_2$, balancing high-level semantic

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information and low-level details [17]. Notably, all three $\mathcal{D}_s^i$ $(i = 1, 2, 3)$ filters first include an identical filter to process $\boldsymbol{s}_s$ with upsampling and downsampling operations incorporated into $\mathcal{D}_s^1$ and $\mathcal{D}_s^3$, respectively, to match the dimensions of the PSCs ($\boldsymbol{s}_p^1$ and $\boldsymbol{s}_p^3$) and feature maps ($\boldsymbol{x}_1$ and $\boldsymbol{x}_3$) in (1) and (3). Upon deriving the SSC and PSCs, we integrate them into a fused SC to enhance the model's feature representation capability and robustness

$$\boldsymbol{s}_f = \mathcal{C}_s \otimes \boldsymbol{s}_s + \mathcal{C}_p^1 \otimes \boldsymbol{s}_p^1 + \mathcal{C}_p^2 \otimes \boldsymbol{s}_p^2 + \mathcal{C}_p^3 \otimes \boldsymbol{s}_p^3 \tag{4}$$

where $\boldsymbol{s}_f \in \mathbb{R}^{P\times h_2\times w_2}$ represents the fused SC, $\mathcal{C}_s \in \mathbb{R}^{P\times P\times k\times k}$ is the fused filter for the SSC, and $\mathcal{C}_p^i \in \mathbb{R}^{P\times P\times k\times k}$ $(i = 1, 2, 3)$ serves as the fused filters for the PSCs. In this work, we replace all manually defined filters with their corresponding convolutional layers. To realize convolutional sparse coding and derive the fused SC, our objective is to freeze the convolutional layers when optimizing the following objective function for both SSC and PSCs:

$$\begin{aligned}\min_{\boldsymbol{s}_s,\boldsymbol{s}_p^1,\boldsymbol{s}_p^2,\boldsymbol{s}_p^3} & \frac{1}{2}\left\|\boldsymbol{x}_1 - \mathcal{D}_s^1 \otimes \boldsymbol{s}_s - \mathcal{D}_p^1 \otimes \boldsymbol{s}_p^1\right\|_F^2 \\ & + \frac{1}{2}\left\|\boldsymbol{x}_2 - \mathcal{D}_s^2 \otimes \boldsymbol{s}_s - \mathcal{D}_p^2 \otimes \boldsymbol{s}_p^2\right\|_F^2 \\ & + \frac{1}{2}\left\|\boldsymbol{x}_3 - \mathcal{D}_s^3 \otimes \boldsymbol{s}_s - \mathcal{D}_p^3 \otimes \boldsymbol{s}_p^3\right\|_F^2 \\ & + \lambda\left(\|\boldsymbol{s}_s\|_{2,1} + \left\|\boldsymbol{s}_p^1\right\|_{2,1} + \left\|\boldsymbol{s}_p^2\right\|_{2,1} + \left\|\boldsymbol{s}_p^3\right\|_{2,1}\right)\end{aligned} \tag{5}$$

where λ serves as a regularization parameter to control the sparsity of the SCs. To enrich spatial information within the SCs, we impose a structured row-sparsity prior constraint on them, i.e., $L_{2,1}$ norm [30], based on the assumption that the same sparsity support set exists among neighboring pixels [19]. This is due to the fact that in HSIs, a spatial neighborhood typically consists of the same material, resulting in highly similar spectral characteristics for neighboring pixels. Consequently, the coefficient features corresponding to these neighboring pixels share the same sparse pattern. It is important to note that the objective function outlined in (5) involves multiple variables requiring optimization. Despite the complexity and challenge of directly optimizing (5), each individual subproblem within the overall equation remains convex. To tackle this multivariable optimization challenge, we employ ADMM [31], which is a robust approach particularly well suited for such problems.

Furthermore, by introducing both AVs and DVs, we define the augmented lagrangian function corresponding to (5)

$$\begin{aligned}& L\left(\boldsymbol{s}_s, \boldsymbol{s}_p^1, \boldsymbol{s}_p^2, \boldsymbol{s}_p^3, \boldsymbol{z}_s, \boldsymbol{z}_p^1, \boldsymbol{z}_p^2, \boldsymbol{z}_p^3, \boldsymbol{\mu}_s, \boldsymbol{\mu}_p^1, \boldsymbol{\mu}_p^2, \boldsymbol{\mu}_p^3\right) \\ & = \frac{1}{2}\sum_{i=1}^{3}\left\|\boldsymbol{x}_i - \mathcal{D}_s^i \otimes \boldsymbol{s}_s - \mathcal{D}_p^i \otimes \boldsymbol{s}_p^i\right\|_F^2 \\ & \quad + \lambda\left(\|\boldsymbol{z}_s\|_{2,1} + \left\|\boldsymbol{z}_p^1\right\|_{2,1} + \left\|\boldsymbol{z}_p^2\right\|_{2,1} + \left\|\boldsymbol{z}_p^3\right\|_{2,1}\right) \\ & \quad + \left\langle\boldsymbol{\mu}_s, \boldsymbol{s}_s - \boldsymbol{z}_s\right\rangle_F + \frac{\rho}{2}\|\boldsymbol{s}_s - \boldsymbol{z}_s\|_F^2 \\ & \quad + \sum_{i=1}^{3}\left[\left\langle\boldsymbol{\mu}_p^i, \boldsymbol{s}_p^i - \boldsymbol{z}_p^i\right\rangle_F + \frac{\rho}{2}\left\|\boldsymbol{s}_p^i - \boldsymbol{z}_p^i\right\|_F^2\right]\end{aligned} \tag{6}$$

where ρ is the penalty parameters that control the consistency of the constraints, $\boldsymbol{z}_s \in \mathbb{R}^{P\times h_2\times w_2}$ and $\boldsymbol{z}_p^i \in \mathbb{R}^{P\times h_i\times w_i}$ are the AVs for the SSC and PSCs, respectively, and $\boldsymbol{\mu}_s \in \mathbb{R}^{P\times h_2\times w_2}$ and $\boldsymbol{\mu}_p^i \in \mathbb{R}^{P\times h_i\times w_i}$ $(i = 1, 2, 3)$ are the DVs for the SSC and PSCs, respectively. Here, F denotes the Frobenius norm extended to the third-order tensors. To optimize (6), we employ ADMM and decompose the equation into six tractable subproblems. We outline three key update procedures: namely, updating $\boldsymbol{s}$, updating $\boldsymbol{z}$, and updating $\boldsymbol{\mu}$. The specifics of the iterative update process are given as follows.

1) Updating $\boldsymbol{s}$: To begin, we isolate the subproblem pertaining to the PSCs from (6)

$$\begin{aligned}\boldsymbol{s}_p^{i,k+1} = \arg\min_{\boldsymbol{s}_p^{i,k}} & \frac{1}{2}\left\|\boldsymbol{x}_i - \mathcal{D}_s^i \otimes \boldsymbol{s}_s^k - \mathcal{D}_p^i \otimes \boldsymbol{s}_p^{i,k}\right\|_F^2 \\ & + \frac{\rho}{2}\left\|\boldsymbol{s}_p^{i,k} - \boldsymbol{z}_p^{i,k} + \boldsymbol{\mu}_p^{i,k}/\rho\right\|_F^2, \quad i = 1, 2, 3\end{aligned} \tag{7}$$

where k indicates the kth iteration. Then, we designate the functional form of (7) as $f(\boldsymbol{s}_p^{i,k})$ and employ the gradient descent algorithm for updating the PSCs

$$\begin{aligned}\nabla_{\boldsymbol{s}_p^{i,k}} f(\boldsymbol{s}_p^{i,k}) = & \left(\mathcal{D}_p^i\right)^T \otimes \left(\mathcal{D}_s^i \otimes \boldsymbol{s}_s^k + \mathcal{D}_p^i \otimes \boldsymbol{s}_p^{i,k} - \boldsymbol{x}_i\right) \\ & + \boldsymbol{\mu}_p^{i,k} + \rho\left(\boldsymbol{s}_p^{i,k} - \boldsymbol{z}_p^{i,k}\right), \quad i = 1, 2, 3\end{aligned} \tag{8}$$

$$\boldsymbol{s}_p^{i,k+1} \leftarrow \boldsymbol{s}_p^{i,k} - \alpha\nabla_{\boldsymbol{s}_p^{i,k}} f(\boldsymbol{s}_p^{i,k}), \quad i = 1, 2, 3 \tag{9}$$

where α is a step size parameter to adjust the update magnitude. The term $(\mathcal{D}_p^i)^T \in \mathbb{R}^{P\times C\times k\times k}$ represents the transposed convolutional operation. To align with the mathematical model and expedite the convergence process, we initialize the transposed convolution layer $(\mathcal{D}_p^i)^T$ by flipping the weights of $\mathcal{D}_p^i$ across the spatial dimensions.

Following the update of the PSCs, we proceed to solve the corresponding subproblem for the SSC:

$$\begin{aligned}\boldsymbol{s}_s^{k+1} = \arg\min_{\boldsymbol{s}_s^k} & \frac{1}{2}\sum_{i=1}^{3}\left\|\boldsymbol{x}_i - \mathcal{D}_s^i \otimes \boldsymbol{s}_s^k - \mathcal{D}_p^i \otimes \boldsymbol{s}_p^{i,k+1}\right\|_F^2 \\ & + \frac{\rho}{2}\left\|\boldsymbol{s}_s^k - \boldsymbol{z}_s^k + \boldsymbol{\mu}_s^k/\rho\right\|_F^2.\end{aligned} \tag{10}$$

The functional form of (10) is defined as $h(\boldsymbol{s}_s^k)$. Next, we obtain

$$\begin{aligned}\nabla_{\boldsymbol{s}_s^k} h(\boldsymbol{s}_s^k) = & \sum_{i=1}^{3}\left[\left(\mathcal{D}_s^i\right)^T \otimes \left(\mathcal{D}_s^i \otimes \boldsymbol{s}_s^k + \mathcal{D}_p^i \otimes \boldsymbol{s}_p^{i,k+1} - \boldsymbol{x}_i\right)\right] \\ & + \boldsymbol{\mu}_s^k + \rho\left(\boldsymbol{s}_s^k - \boldsymbol{z}_s^k\right)\end{aligned} \tag{11}$$

$$\boldsymbol{s}_s^{k+1} \leftarrow \boldsymbol{s}_s^k - \alpha\nabla_{\boldsymbol{s}_s^k} h\left(\boldsymbol{s}_s^k\right). \tag{12}$$

Similarly, $(\mathcal{D}_s^i)^T \in \mathbb{R}^{P\times C\times k\times k}$ $(i = 1, 2, 3)$ serves as the transposed convolutional layer corresponding to $\mathcal{D}_s^i$. It is derived by flipping $\mathcal{D}_s^i$ along its spatial dimensions.

2) Updating $\boldsymbol{z}$: To update the AVs for the PSCs, the subproblem can be written as follows:

$$\begin{aligned}\boldsymbol{z}_p^{i,k+1} = \arg\min_{\boldsymbol{z}_p^{i,k}} & \frac{\rho}{2}\left\|\boldsymbol{s}_p^{i,k+1} - \boldsymbol{z}_p^{i,k} + \boldsymbol{\mu}_p^{i,k}/\rho\right\|_F^2 \\ & + \lambda\left\|\boldsymbol{z}_p^{i,k}\right\|_{2,1}, \quad i = 1, 2, 3.\end{aligned} \tag{13}$$

Equation (13) is traditionally solved using the well-known soft-thresholding operation, $\mathcal{S}_{(\lambda/\rho)}(\boldsymbol{a}) = \text{sign}(\boldsymbol{a})\max(|\boldsymbol{a}| - \lambda/\rho, 0)$ [31]. However, this elementwise operation fails to

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incorporate spatial information. In contrast, (13) leverages the $L_{2,1}$ norm to introduce structured constraints, enriching the solution space with spatial information. This makes the standard soft-thresholding operation ill-suited for our purposes. Therefore, to meet the row-sparsity requirement, we employ the following group soft-thresholding method [32]:

$$\text{soft}_{\text{group}}(\boldsymbol{v}_j, \xi) = \max\left(1 - \frac{\xi}{\|\boldsymbol{v}_j\|_2}, 0\right)\boldsymbol{v}_j, \quad j = 1, 2, \ldots, P. \tag{14}$$

Here, $\boldsymbol{v} \in \mathbb{R}^{P\times N}$, j represents the jth row, and ξ is a predefined threshold value. N represents the number of pixels. This approach imposes structured constraints on the input while appropriately scaling it, enhancing the compactness and effectiveness of the coefficient features. Specifically, when the L_2 norm of a row exceeds a threshold, the information in that feature channel is further compressed, selectively retaining the most important spatial location information. Conversely, when the norm is below the threshold, indicating relative redundancy, the information is removed to prevent interference, effectively reducing spectral redundancy and benefiting subsequent detection. By applying (14) to (13), we have

$$\boldsymbol{z}_p^{i,k+1} \leftarrow \left[\max\left(1 - \frac{\lambda/\rho}{\left\|\left(\boldsymbol{s}_p^{i,k+1} + \boldsymbol{\mu}_p^{i,k}/\rho\right)_j\right\|_2}, 0\right) \cdot \left(\boldsymbol{s}_p^{i,k+1} + \boldsymbol{\mu}_p^{i,k}/\rho\right)_j\right]_{j=1}^{P}, \quad i = 1, 2, 3. \tag{15}$$

Notably, the term $\boldsymbol{s}_p^{i,(k+1)} + \boldsymbol{\mu}_p^{i,k}/\rho \in \mathbb{R}^{P\times h_i\times w_i}$ must be reshaped into matrix form prior to its application in (15). After obtaining the updated results, these matrices are transformed back into 3-D tensors. This step is crucial for preserving the integrity of the spatial information.

Likewise, we have the following subproblem for $\boldsymbol{z}_s$:

$$\boldsymbol{z}_s^{k+1} = \arg\min_{\boldsymbol{z}_s^k} \frac{\rho}{2}\left\|\boldsymbol{s}_s^{k+1} - \boldsymbol{z}_s^k + \boldsymbol{\mu}_s^k/\rho\right\|_F^2 + \lambda\left\|\boldsymbol{z}_s^k\right\|_{2,1}. \tag{16}$$

By employing the group soft-thresholding operation on the given subproblem, we can derive the following update strategy for $\boldsymbol{z}_s$:

$$\boldsymbol{z}_s^{k+1} \leftarrow \left[\max\left(1 - \frac{\lambda/\rho}{\left\|\left(\boldsymbol{s}_s^{k+1} + \boldsymbol{\mu}_s^k/\rho\right)_j\right\|_2}, 0\right) \cdot \left(\boldsymbol{s}_s^{k+1} + \boldsymbol{\mu}_s^k/\rho\right)_j\right]_{j=1}^{P}. \tag{17}$$

3) Updating $\boldsymbol{\mu}$: Next, we proceed to update the DVs for the PSCs and the SSC using the following two distinct approaches:

$$\boldsymbol{\mu}_p^{i,k+1} \leftarrow \boldsymbol{\mu}_p^{i,k} + \rho\left(\boldsymbol{s}_p^{i,k+1} - \boldsymbol{z}_p^{i,k+1}\right), \quad i = 1, 2, 3 \tag{18}$$

$$\boldsymbol{\mu}_s^{k+1} \leftarrow \boldsymbol{\mu}_s^k + \rho\left(\boldsymbol{s}_s^{k+1} - \boldsymbol{z}_s^{k+1}\right). \tag{19}$$

After $K+1$ iterations using the aforementioned update strategies, we combine the obtained SSC and PSCs through (4) to derive the fused SC.

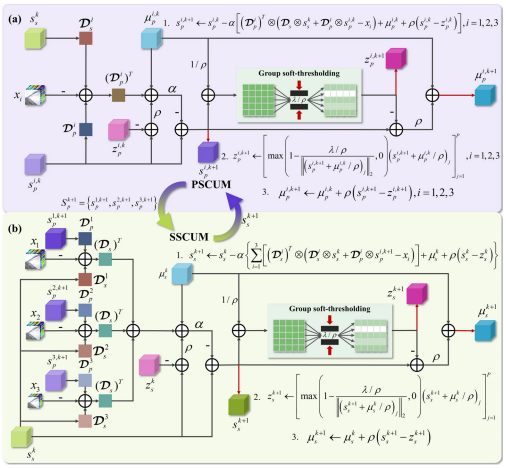


Fig. 3. Detailed architectures of PSCUM and SSCUM in the $(k+1)$th iteration. (a) PSCUM. (b) SSCUM.

B. Multiscale Convolutional Sparse Coding-Guided Deep Interpretable Network

In light of the well-formulated mathematical model, we extend its capabilities by unfolding it into a deep neural network with $K+1$ iterations. This results in a transparent, interpretable encoder architecture, MSCSCNet. Our network leverages the computational power and automated parameter tuning of DL, overcoming the model's limitation of capturing only shallow features. The specific network architecture is illustrated in Fig. 1(a).

Benefiting from the MSCSCNet, the PVs and SVs are updated during forward propagation with deep feature embedding. The PVs include $\boldsymbol{S}_p$ (with $\boldsymbol{s}_p^1$, $\boldsymbol{s}_p^2$, and $\boldsymbol{s}_p^3$), $\boldsymbol{z}_p^1$, $\boldsymbol{\mu}_p^1$, $\boldsymbol{z}_p^2$, $\boldsymbol{\mu}_p^2$, $\boldsymbol{z}_p^3$, and $\boldsymbol{\mu}_p^3$, while the SVs include $\boldsymbol{s}_s$, $\boldsymbol{z}_s$, and $\boldsymbol{\mu}_s$. The network features two submodules: PSCUM and SSCUM. The PSCUM is constructed in accordance with (9), (15), and (18), while the SSCUM follows (12), (17), and (19). The input patch $\boldsymbol{x}$ is processed through convolutional layers and downsampling, producing feature maps at three scales, denoted as $X = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$. Residual connections [33] with convolutional downsampling are used to mitigate information loss. Subsequently, we initialize the PVs and SVs using X, and the iterative update process alternates between PSCUM and SSCUM for $K+1$ iterations. The final $\boldsymbol{s}_f$ is obtained by fusing the resulting SSC and PSCs through (4).

1) Private Sparse Coefficient Update Module: As delineated in Fig. 3(a), the PSCUM is employed for updating the PVs. To harness the strengths of DL, the operators $\boldsymbol{\mathcal{D}}_s^i$, $\boldsymbol{\mathcal{D}}_p^i$, and $(\boldsymbol{\mathcal{D}}_p^i)^T$ $(i = 1, 2, 3)$ are implemented using learnable convolutional layers. Through (9), the PSCs and SSC from the previous iteration are processed by their respective convolutional layers, remapping the private and shared coefficient features back to the feature space to compute differences with the corresponding feature maps. These differences are then reprojected into the private coefficient spaces using transposed convolutional layers. The regularization terms $\boldsymbol{\mu}_p^{i,k} + \rho(\boldsymbol{s}_p^{i,k} - \boldsymbol{z}_p^{i,k})$ are applied to maintain and enhance sparsity. As a result, the updated PSCs capture private structural information at different scales and reduce redundancy. The AVs are then updated using the activation function from (14), which applies structured sparse constraints

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to enhance sparsity of the next PSCs. Finally, the DVs are updated following (18), adjusting the optimization direction and magnitude for the next PSC update, gradually eliminating the bias between the PSCs and AVs.

2) Shared Sparse Coefficient Update Module: The SSCUM is responsible for updating the SVs, as specifically outlined in Fig. 3(b). Similarly, through (12), the previous SSC and updated PSCs are remapped to the respective feature spaces at different scales using convolutional layers to compute differences with corresponding feature maps. These differences are reprojected into the shared coefficient space by the transposed convolutional layer and then aggregated to form a comprehensive and shared difference. Then, the regularization terms $\boldsymbol{\mu}_s^k + \rho(\boldsymbol{s}_s^k - \boldsymbol{z}_s^k)$ are also applied to improve sparsity. Relative to the PSCs, the updated SSC captures recurring textural and structural patterns that are consistent across scales. Following this, a group soft-thresholding operation is then applied to produce the updated AV, imposing structured sparse constraints and promoting sparsity for the next SSC. Finally, (19) is exploited to update the DV, adjusting the optimization process for the subsequent SSC update and gradually reducing the bias between the SSC and AV. Furthermore, λ, α, and ρ are key parameters controlling sparsity, update magnitude, and constraint consistency, respectively. λ influences feature selection and mitigates overfitting; α ensures a balance between convergence speed and stability, and ρ guarantees that the SCs meet their constraints. These parameters are learnable, automatically adjusting during training to reduce reliance on manual tuning.

In this way, the internal mechanisms and information flow of PSCUM and SSCUM in MSCSCNet are transparent and traceable, with each operation aligned with the derived iterative algorithms. By alternating PSCUM and SSCUM, the network optimizes PSCs and SSC during forward propagation and algorithm parameters through gradient backpropagation, merging the strengths of model-based and DL approaches. Equation (4) then aggregates these coefficients to produce robust, low-redundancy coefficient features. This contrasts with conventional data-driven deep neural networks, which often rely on stacking subjectively designed modules and various feature extractors, lacking mathematical model guidance and transparent internal mechanisms, making them difficult to understand and dependent on extensive trial-and-error.

C. Two-Stage Semi-Supervised Training Strategy

As illustrated in Fig. 1(b), we introduce a two-stage semi-supervised training strategy for MSCSCNet. At first, SL is exclusively deployed for HSI-CD using limited labeled data. Focal loss guides the network optimization to alleviate the imbalance between changed and unchanged classes [34]

$$\mathcal{L}_{\mathrm{sl}} = -\frac{1}{B}\sum_{b=1}^{B} \alpha_f^b \left(1 - p_f^b\right)^{\gamma_f} \log\left(p_f^b\right) \tag{20}$$

$$\alpha_f^b = \begin{cases} \tilde{\alpha}, & \text{if } y^b = 1 \\ 1 - \tilde{\alpha}, & \text{otherwise} \end{cases} \tag{21}$$

$$p_f^b = \begin{cases} p^b, & \text{if } y^b = 1 \\ 1 - p^b, & \text{otherwise} \end{cases} \tag{22}$$

where B denotes the batch size, y^b represents the ground-truth label for the bth sample, α_f^b is a weighting factor for class imbalance, and p^b is the predicted probability. In addition, the focusing parameter γ_f and the modulating factor $(1 - p_f^b)^{\gamma_f}$ reduce loss contribution from easily classified samples. We set $\tilde{\alpha} = 0.25$ and $\gamma_f = 2$ as per [34]. In this manner, the MSCSCNet undergoes an effective "warm-up" phase, ensuring robust initialization for the subsequent training stage.

In Fig. 1(b), the CS-TSSD features student and teacher branches, each containing siamese subbranches. Both branches utilize dual MSCSCNet-projection head pathways with shared weights. The projection head is a three-layer MLP, followed by L_2-norm normalization and a weight normalization layer [35]. The student's weights are updated via gradient backpropagation, while the teacher's weights are updated using an EMA of the student's, stopping its gradient [35]

$$\theta_t \leftarrow \beta\theta_t + (1 - \beta)\theta_s. \tag{23}$$

Here, θ_t represents the teacher's weights, θ_s denotes the student's weights, and β acts as a smoothing factor controlling the update speed. The teacher's weights are a running average of the student's, stabilizing the teacher by mitigating the student's stochastic variations. This stability ensures that the teacher consistently outperforms the student, offering effective guidance. In light of our smaller dataset and network architecture, a lower β of 0.917 is set to accelerate the teacher's learning, which ensures that the teacher is effectively smoothed by historical information while remaining adaptable to new information. Furthermore, to prevent training collapse, we employ centering and sharpening operations in line with self-distillation with no labels (DINO) [35].

We feed a substantial volume of unlabeled bitemporal patches into the student and teacher branches. Subsequently, a hybrid training strategy for MSCSCNet is then implemented, combining both SL and CS-TSSD methods. The composite loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{sl}} + \eta\mathcal{L}_{\mathrm{ssl}}. \tag{24}$$

Here, the tradeoff parameter η controls the contribution of $\mathcal{L}_{\mathrm{ssl}}$, which is derived from the individual losses $\mathcal{L}_u$ and $\mathcal{L}_d$ as follows:

$$\mathcal{L}_{\mathrm{ssl}} = \psi\mathcal{L}_u + (1 - \psi)\mathcal{L}_d. \tag{25}$$

First, upon acquiring the learned representations from each branch, we construct a symmetric self-supervised loss function as $\mathcal{L}_u$ [35]

$$\mathcal{L}_u = -\frac{1}{B}\sum_{b=1}^{B} \frac{1}{2}\left(\boldsymbol{f}_{t,2}^b \log \boldsymbol{f}_{s,1}^b + \boldsymbol{f}_{t,1}^b \log \boldsymbol{f}_{s,2}^b\right) \tag{26}$$

where $\boldsymbol{f}_{s,1}^b \in \mathbb{R}^O$ and $\boldsymbol{f}_{s,2}^b \in \mathbb{R}^O$ are the bth student's learned representations from bitemporal patches, and $\boldsymbol{f}_{t,1}^b \in \mathbb{R}^O$ and $\boldsymbol{f}_{t,2}^b \in \mathbb{R}^O$ denote that those from the teacher O are the output dimensions of the projection head. This loss encourages the student's representations to converge toward the teacher's, which is effective in static scenarios. In dynamic environments, aligning the student's features directly with the teacher's may cause the network to overlook important change components

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vital for CD. To address this, we compute the difference between the learned representation pairs of the student and teacher, encouraging the student's difference representation to approximate the teacher's, thereby remaining sensitive to essential change components

$$\mathcal{L}_d = -\frac{1}{B}\sum_{b=1}^{B} \boldsymbol{d}_t^b \log \boldsymbol{d}_s^b \tag{27}$$

where $\boldsymbol{d}_s^b \in \mathbb{R}^O$ and $\boldsymbol{d}_t^b \in \mathbb{R}^O$ are the bth difference representations of the student and teacher, respectively. To adapt them to different scenarios, we leverage the teacher's superior performance and stability during training [35]. The cosine similarity between the teacher's learned representations is calculated to derive a dynamic weight, which modulates the loss terms $\mathcal{L}_u$ and $\mathcal{L}_d$. The formulation of the dynamic weight is as follows:

$$\psi = \left(\frac{1+\text{sim}(\boldsymbol{f}_{t,1},\boldsymbol{f}_{t,2})}{2}\right)^{\gamma} \tag{28}$$

where $\text{sim}(\cdot,\cdot)$ represents the cosine similarity and γ serves as a penalty factor. By setting $\gamma > 1$, the weight attributed to $\mathcal{L}_u$ diminishes when the bitemporal patches exhibit dissimilarity. This renders the network more sensitive to changes, which is significant to the CD task.

In addition to CS-TSSD, we also employ SL on the student with a few labeled patches. This dual approach leverages both unlabeled and labeled data, ensuring accurate weight updates and enhancing MSCSCNet's performance. During inference, only the well-trained MSCSCNet from the student and the classifier are used for HSI-CD.

III. Experimental Results and Analysis

In this section, we validate the effectiveness of the proposed method. First, we introduce the benchmark datasets and evaluation metrics used. Next, we detail the competing methods and experimental settings. Following this, we compare the performance of our method with the competing methods across different datasets and conduct extensive ablation studies on different components of the proposed method. Finally, we discuss the strengths and limitations of the proposed method.

A. Datasets and Evaluation Metrics

We conduct extensive experiments on three public benchmark HSI-CD datasets (see Fig. 4) to demonstrate the effectiveness of the proposed method, namely, Farmland dataset,[1] River dataset,[2] and Santa Barbara dataset.[3] The first two datasets were captured by the Earth Observing-1 (EO-1) Hyperion sensor [36] with a 30-m spatial resolution, while the latter was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with a 20-m spatial resolution. All datasets span a spectral range of 0.4–2.5 μm. Detailed information on these datasets is presented in Table I.

[1]https://rslab.ut.ac.ir/data

[2]https://pan.baidu.com/s/14ht8k5H-8ObzHJS6msYZjQ

[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detectiondataset

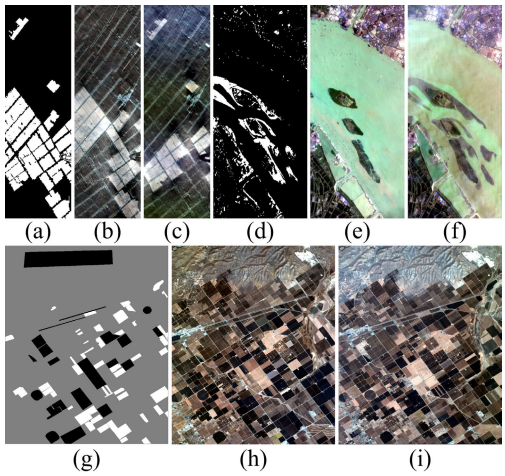


Fig. 4. Farmland, River, and Santa Barbara datasets. (a) Ground truth of the Farmland dataset. (b) and (c) Farmland pseudocolor images acquired at T_1 and T_2, respectively. (d) Ground truth of the River dataset. (e) and (f) River pseudocolor images acquired at T_1 and T_2, respectively. (g) Ground truth of the Santa Barbara dataset. (h) and (i) Santa Barbara pseudocolor images acquired at T_1 and T_2, respectively. In the ground-truth maps, white and black denote changed and unchanged pixels, respectively.

TABLE I
Detailed Information on the Three Datasets

Dataset	Farmland	River	Santa Barbara
Date T_1	May 3, 2006	May 3, 2013	2013
Date T_2	Apr. 23, 2007	Dec. 31, 2013	2014
Image Size	420×140	463×241	984×740
Spectral Bands	154	198	224
Unchanged Pixels	40,417	101,885	80,418
Changed Pixels	18,383	9,698	52,134

In order to quantitatively and objectively assess the performance of the proposed method, we primarily adopt four evaluation metrics: overall accuracy (OA), kappa coefficient (KC), F1 score (F1), and average accuracy (AA). Prior to computing them, we need to count the number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN). In this context, TP and TN refer to the changed and unchanged pixels that are correctly detected, respectively. FP denotes the misclassification of unchanged pixels as changed, while FN is the opposite. In visualization results, white, black, red, green, and gray denote TP, TN, FP, FN, and undetermined pixels, respectively.

B. Competing Methods and Experimental Settings

To demonstrate the effectiveness of our method, we compared it with eight representative methods. Three classic CD methods were implemented, including CVA [4], IR-MAD [7], and support vector machine (SVM) [37]. In addition, we included five state-of-the-art DL methods: D^{2}AGCN [38], which uses a dual-branch difference amplification graph convolutional network for HSI-CD; D-LIEG [39], which employs a dual-branch local information-enhanced graph transformer to exploit local–global spectral–spatial features with limited training samples; MSDFFN [17], which adopts a multiscale diff-changed feature fusion strategy; DA-Former

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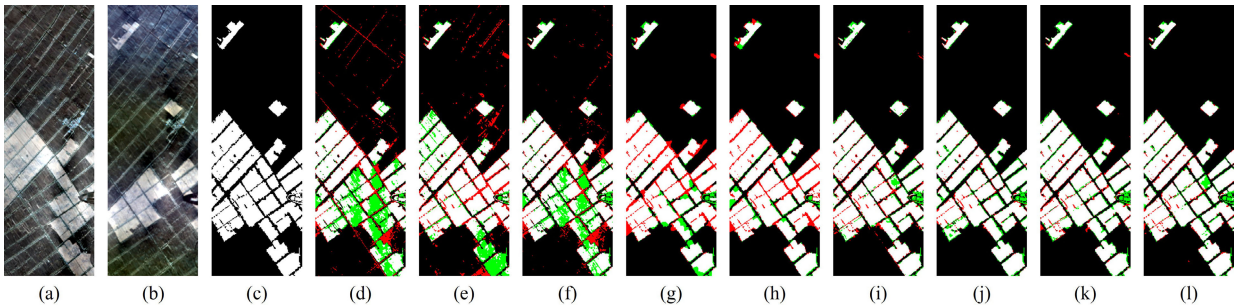


Fig. 5. Visualization results of various methods on the Farmland dataset. (a) Pseudocolor image at T_1. (b) Pseudocolor image at T_2. (c) Ground truth. (d) CVA. (e) IR-MAD. (f) SVM. (g) D-LIEG. (h) D^{2}AGCN. (i) MSDFFN. (j) DA-Former. (k) MFECN. (l) Our proposed method.

[40], a semi-supervised approach using a domain alignment transformer; and multi-level features fused and change information enhanced neural network (MFCEN) [41], which uses a siamese-based network and a temporal change information branch for hierarchical feature extraction and multilevel aggregation. We trained the DL methods, including our proposed method, using 1% of labeled samples from each class, except for the unsupervised CVA and IR-MAD. In addition, we combined the labeled samples with randomly selected unlabeled samples for training DA-Former and our method. All DL methods were reimplemented, with hyperparameters set according to their original papers.

Taking the preservation of spatial information and computation efficiency into consideration, we take as inputs overlapping patch pairs with a size of 11 × 11. Specifically, an 11 × 11 patch is generated by centering on a single pixel and including its neighboring pixels, with the label of the patch corresponding to the label of the central pixel. Consequently, after a series of convolutional downsampling and residual connections, the resulting three feature maps have scales of 9, 7, and 5. The proposed method is trained and tested on the PyTorch framework using an NVIDIA GeForce RTX 3090 GPU. Batch sizes for labeled and unlabeled samples are set to 32 and 512, respectively. The training encompasses 100 epochs: 15 for supervised initialization and 85 for a hybrid training of SL and CS-TSSD. We use an stochastic gradient descent (SGD) optimizer with a weight decay of $5e^{-3}$, a momentum of 0.9, and a learning rate that ramps up to 0.032 in the first 5 epochs and then decays using a cosine schedule. In addition, the hidden feature P of the MSCSCNet is set to 256, and the number of forward iterative updates is 3. The projection head includes three linear layers with a hidden dimension of 2048 and an output dimension of 256, using Gaussian error linear unit (GELU) as the activation function. The smoothing factor β is set to a small value of 0.917, considering the relatively small scale of the dataset and network architecture. The penalty factor γ and the tradeoff parameter η are set to 1.6 and 0.7, respectively. The random seed is set to 42. Moreover, the remaining hyperparameters are based on DINO [35] and modified to enhance training stability.

C. Experimental Results

1) Experimental Results on the Farmland Dataset: The quantitative assessment of different methods on the Farmland dataset is shown in Table II. CVA and IR-MAD, relying on handcrafted features and statistical models, fail to capture complex land cover changes, resulting in poor performance. SVM performs suboptimally due to limited labeled samples. DL-based methods, such as D-LIEG, D^{2}AGCN, MSDFFN, and MFCEN, show significant improvements. The semi-supervised DA-Former outperforms these supervised DL models by aligning bitemporal images and learning from unlabeled data. Our method achieves the best results across all metrics, demonstrating the effectiveness of the CS-TSSD paradigm and the integration of DL with model-based methods.

TABLE II
QUANTITATIVE ASSESSMENT OF THE PROPOSED METHOD AND OTHER COMPARATIVE METHODS ON THE FARMLAND DATASET

Method	OA%	KC(×100)	F1%	AA%
CVA	87.49	69.98	78.82	83.94
IR-MAD	89.74	76.32	83.82	88.45
SVM	89.80	75.82	83.13	87.23
D-LIEG	92.49	82.88	88.43	92.31
D^{2}AGCN	93.97	86.27	90.73	94.11
MSDFFN	95.16	88.74	92.26	94.36
DA-Former	95.91	90.36	93.31	94.64
MFCEN	95.50	89.47	92.73	94.51
Ours	**95.97**	**90.58**	**93.50**	**95.06**

Fig. 5 shows the visualization results on the Farmland dataset. Traditional CD methods extract shallow features, resulting in higher FP and FN. In contrast, DL algorithms, such as D-LIEG and D^{2}AGCN, significantly improve detection results. MSDFFN, DA-Former, and MFCEN further reduce FP and FN, with DA-Former achieving the second-best results. Our proposed method surpasses DA-Former, particularly in handling complex or subtle changes in areas with indistinct boundaries. However, it is observed that, compared with DA-Former and MFCEN, our method and MSDFFN both show an increase in FN in the specific patch areas. This may be attributed to limited training sample coverage in these areas or the better applicability of the transformer's long-range modeling in these specific local regions.

2) Experimental Results on the River Dataset: As presented in Table III, CVA and IR-MAD show significantly lower performance in OA, KC, and F1, yet they exhibit higher AA values, suggesting a notable detection imbalance. The supervised SVM achieves improved performance in OA but lower AA values. D-LIEG and D^{2}AGCN have moderate results, possibly due to the presence of multiple classes

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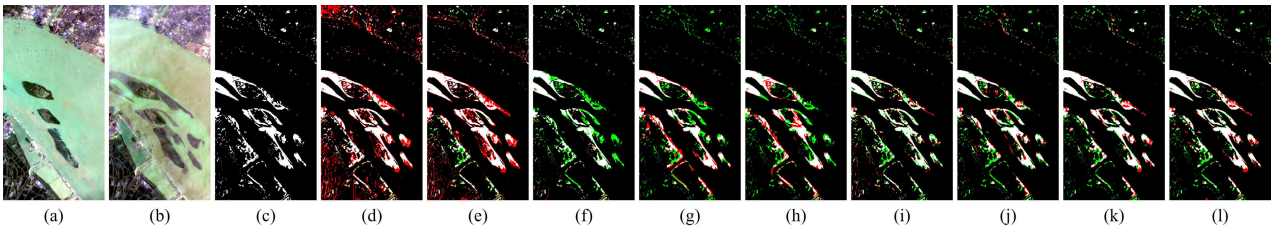


Fig. 6. Visualization results of various methods on the River dataset. (a) Pseudocolor image at T_1. (b) Pseudocolor image at T_2. (c) Ground truth. (d) CVA. (e) IR-MAD. (f) SVM. (g) D-LIEG. (h) D^{2}AGCN. (i) MSDFFN. (j) DA-Former. (k) MFECN. (l) Our proposed method.

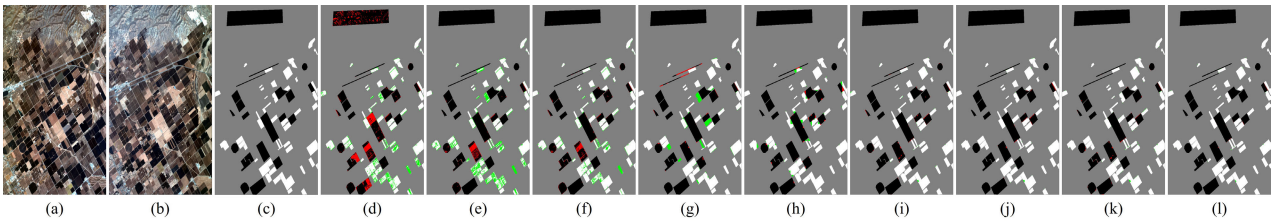


Fig. 7. Visualization results of various methods on the Santa Barbara dataset. (a) Pseudocolor image at T_1. (b) Pseudocolor image at T_2. (c) Ground truth. (d) CVA. (e) IR-MAD. (f) SVM. (g) D-LIEG. (h) D^{2}AGCN. (i) MSDFFN. (j) DA-Former. (k) MFECN. (l) Our proposed method.

within their segmented superpixels, particularly in datasets like River with dispersed and fragmented change areas. The transformer-based DA-Former underperforms compared with MSDFFN. However, MFCEN, which also uses a transformer but leverages multilevel and enhanced change features, outperforms MSDFFN in overall performance. This suggests that vanilla transformer's long-range modeling is less effective for scattered, granular datasets. In contrast, our method, with multiscale feature extraction and sparse coding, achieves superior results.

The visualization results in Fig. 6 show CVA and IR-MAD with poor outcomes, much higher FP than FN, but high AA values. This reveals the limitations of the AA metric, which overlooks FP and FN differences and fails to reflect overall model performance in imbalanced datasets. Conversely, SVM shows a higher FN than FP. D-LIEG and D^{2}AGCN show improved balance but still suffer from many misclassifications. DA-Former, MFCEN, MSDFFN, and our method show significant improvements, with MSDFFN and our method exceling at detecting subtle changes. Our method further reduces noise in large continuous change areas due to its sparse coding design, effectively mitigating noise and redundant information. Notably, despite better overall performance, our method shows increased FP in certain areas. This phenomenon indicates that while the introduction of CS-TSSD increases the network's sensitivity to changes, it may be overly sensitive in the specific areas, resulting in the misclassification of these unchanged regions.

3) Experimental Results on the Santa Barbara Dataset: Table IV displays the quantitative assessment of different methods on the Santa Barbara dataset. Among traditional methods, unsupervised CVA and IR-MAD perform the worst, while supervised SVM shows improvements. DL-based D-LIEG and D^{2}AGCN further enhance CD results. Moreover, MSDFFN, MFCEN, and DA-Former perform better, with semi-supervised DA-Former outperforming supervised MSDFFN and MFCEN with limited labeled data.

TABLE III

QUANTITATIVE ASSESSMENT OF THE PROPOSED METHOD AND OTHER COMPARATIVE METHODS ON THE RIVER DATASET

Method	OA%	KC(×100)	F1%	AA%
CVA	92.67	65.75	69.55	**94.30**
IR-MAD	93.82	67.98	71.29	91.30
SVM	95.65	65.99	68.14	76.61
D-LIEG	94.30	61.67	64.74	78.86
D^{2}AGCN	94.15	60.32	63.48	78.02
MSDFFN	95.96	72.87	75.06	84.21
DA-Former	95.46	69.08	71.53	81.98
MFCEN	95.89	73.07	75.31	85.13
Ours	**96.07**	**74.24**	**76.37**	85.64

TABLE IV

QUANTITATIVE ASSESSMENT OF THE PROPOSED METHOD AND OTHER COMPARATIVE METHODS ON THE SANTA BARBARA DATASET

Method	OA%	KC(×100)	F1%	AA%
CVA	83.25	65.19	79.20	82.86
IR-MAD	87.42	72.69	82.27	85.10
SVM	93.15	85.49	91.02	92.29
D-LIEG	94.95	89.30	93.36	94.13
D^{2}AGCN	97.28	94.27	96.50	96.95
MSDFFN	99.04	98.00	98.79	99.07
DA-Former	99.34	98.62	99.16	99.36
MFCEN	98.92	97.74	98.63	98.89
Ours	**99.67**	**99.31**	**99.58**	**99.65**

Our proposed method surpasses all these methods across all metrics, demonstrating the effectiveness of combining DL with model-based approaches.

As shown in Fig. 7, the visualization results of CVA are the poorest. IR-MAD and SVM show relative improvements yet still suffer from significant misclassifications. D-LIEG and D^{2}AGCN exhibit a notable reduction in FP and FN, with the former underperforming the latter due to the limitation of computational overhead. The performance of MSDFFN, MFCEN, and DA-Former is further enhanced, virtually eliminating extensive misclassifications. Moreover, our method

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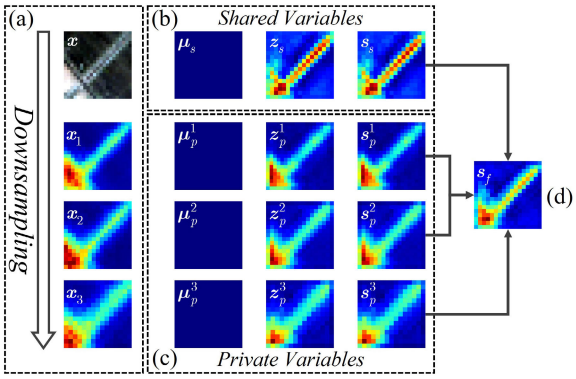


Fig. 8. Feature visualization of the proposed MSCSCNet. (a) Input patch and its feature maps at three different scales. (b) Cross-scale SVs. (c) Scale-specific PVs. (d) Fused SC feature.

demonstrates the best visualization results, reducing noise in continuous areas and refining boundaries.

4) Feature Visualization: To demonstrate the effectiveness of our method, Fig. 8 visualizes its results. Feature maps at different scales from the same input patch show varying levels of texture and structural detail. After optimizing at their respective scales, the redundancy in the PSC features is reduced, resulting in more concentrated feature distributions. Unlike PSC features, the SSC features not only capture but also significantly enhance shared textures and structures across all scales, with reduced redundancy. Moreover, all AVs representing the ideal sparse solution closely resemble the SC features. The small DVs further demonstrate the consistency between them, which is as expected. The fused SC features integrate both PSC and SSC features, enhancing texture and structural information while further reducing redundancy. This validates our method's ability to guide the feature extraction process, yielding goal-oriented coefficient features.

D. Ablation Study

1) Number of Forward Iterative Updates: To assess the impact of different numbers of forward iterative updates on model performance, we experimented with various $K+1$ settings (1, 2, 3, 4, 5) on the three datasets. Fig. 9 illustrates that performance improves as $K+1$ increases from 1 to 3, due to better exploitation of the model's internal update regularization and learning of more complex feature combinations. However, performance deteriorates when $K+1$ exceeds 3, likely due to overfitting from the more complex model. All things considered, we set $K+1$ to 3 in our experiments for an optimal tradeoff between performance and complexity.

2) Group Soft-Thresholding: To verify the effectiveness of the group soft-thresholding method, we conducted a comparison between the soft-thresholding and group soft-thresholding methods on the three datasets, as shown in Fig. 10. We can see that the group soft-thresholding method outperforms the traditional soft-thresholding method on all three datasets. Unlike the elementwise soft-thresholding method, the group method can capture the spatial structural information and scale data more robustly, mitigating the incidence of errors due to single-pixel thresholding. This enhancement significantly reduces FN, thereby improving detection results.

3) Patch Size: As shown in Fig. 11, we conducted experiments on the three datasets to assess the impact of patch size on the performance of the model. Across all datasets, performance increases as the patch size grows from 7 to 11, peaking at a patch size of 11. However, performance declines with further increases in patch size. HSIs typically have high spectral resolution but low spatial resolution. Larger patches can introduce noise and redundant information, reducing local region similarity and increasing computational complexity. Conversely, smaller patches may not capture enough spatial–spectral information. In contrast, a patch size of 11×11 strikes a balance, retaining rich spatial context and spectral information while ensuring local consistency and reasonable computational efficiency.

4) Feature Fusion: We conducted experiments on three datasets to assess the impact of addition, subtraction, and concatenation as feature fusion methods for bitemporal images. As shown in Table V, subtraction (corresponding to "Sub" in Table V) achieves the best results. Addition (corresponding to "Add" in Table V) fails to highlight changes, making it less effective for CD. Although concatenation (corresponding to "Concat" in Table V) retains all information, the increased feature dimensions might introduce redundancy, interfering with detection. In contrast, subtraction captures differences directly and highlights changes, enabling the model to focus more effectively on areas of actual change.

5) Scale of Shared Sparse Coefficient: To evaluate the impact of different scales of SSC on model performance, we tested small (S), middle (M), and large (L) scales (5, 7, and 9, corresponding to feature maps $\boldsymbol{x_3}$, $\boldsymbol{x_2}$, and $\boldsymbol{x_1}$, respectively) across three datasets with input patches of size 11. As shown in Table VI, the middle scale performs best on all datasets. From the perspective of feature maps, the large scale in shallow layers captures low-level details but introduces information redundancy and lacks comprehensive global semantics, reducing robustness. The small scale in deeper layers, while rich in semantics, fails to capture local structures and subtle details. The middle scale balances semantic information and low-level details, leading to more effective detection.

6) PSCUM and SSCUM: To demonstrate the impact of considering both PSCs and SSC simultaneously on the performance of the model, we conducted ablation studies across the three datasets, as presented in Table VII. When PSCUM and SSCUM work together, the model achieves optimal performance. With only PSCUM, the model fails to differentiate between PSCs and SSC, leading to redundant feature extraction across scales, which hinders subsequent detection. Conversely, with only SSCUM, the model relies solely on shared cross-scale features and misses unique details at each scale, resulting in less refined features. By considering both PSCs and SSC, the model can explicitly encode scale-specific and shared features, improving interpretability and extracting more comprehensive and compact coefficient features.

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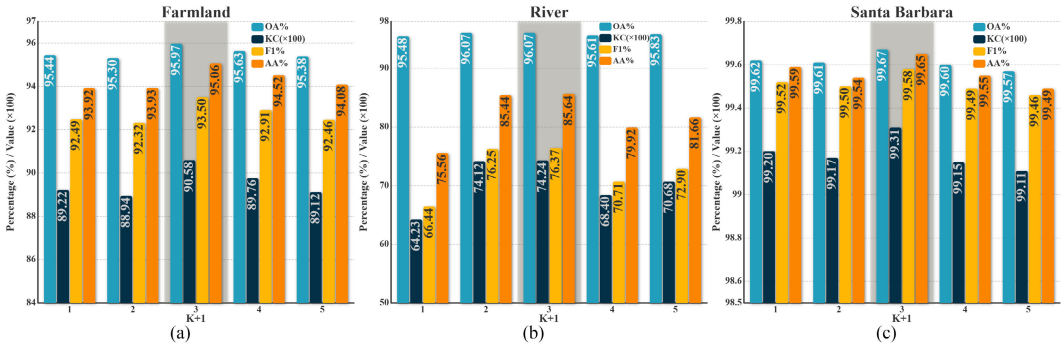


Fig. 9. Experimental results of different numbers of forward iterative update on the three datasets. (a) Farmland dataset. (b) River dataset. (c) Santa Barbara dataset.

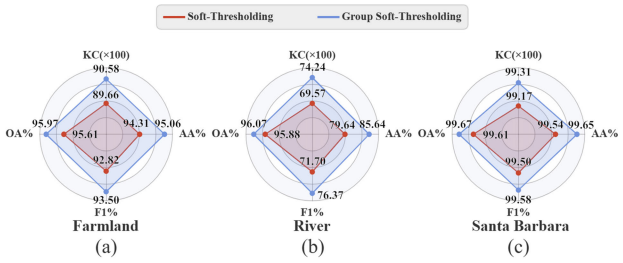


Fig. 10. Comparison of the soft-thresholding and group soft-thresholding methods on three datasets. (a) Farmland dataset. (b) River dataset. (c) Santa Barbara dataset.

7) Change-Sensitive Teacher–Student Self-Distillation : Fig. 12 shows the effect of CS-TSSD and varying η on the OA and KC of the model across three datasets. With $\eta = 0$, indicating the absence of CS-TSSD, the model undergoes only SL. For the Farmland and Santa Barbara datasets, performance generally improves despite oscillations with $\eta > 0$, peaking at $\eta = 0.7$, demonstrating CS-TSSD's effectiveness in utilizing unlabeled data. For the River dataset, performance improves at $\eta \geq 0.4$, peaking at 0.7. However, during $0.1 \leq \eta \leq 0.4$, performance declines, possibly due to insufficient weight on learning from complex unlabeled data, inadvertently hindering the normal learning process from labeled data. When $\eta > 0.7$, performance declines across all datasets, likely due to overreliance on unlabeled data. Therefore, we incorporate CS-TSSD into our model and select $\eta = 0.7$ as the optimal setting.

8) Penalty Factor: Table VIII displays the effect of different γ values on the three datasets. Among six different γ values, the model achieves optimal performance on all datasets when γ is set to 1.6. With $\gamma = 0$, the model focuses only on invariance from unlabeled data, leading to lower detection results. As γ increases, performance on the Farmland and Santa Barbara datasets improves, indicating a gradual enhancement in sensitivity to changes. For the River dataset, performance slightly decreases at $\gamma = 0.4$, suggesting that a low γ may not capture complex changes well, inadvertently squeezing the learning of invariant information. At $\gamma = 1.6$, the model balances sensitivity to changed and invariant components, performing best on all datasets. However, at $\gamma = 2$, performance drops due to excessive sensitivity, causing misclassification of unchanged areas. Thus, we set γ to 1.6 in our experiments.

9) Computational Cost: As illustrated in Table IX, we analyzed the computational costs of all DL-based methods and our method. Although our method has moderate Params and FLOPs, it is the most time-consuming. This may be because our approach combines model-based and DL methods, involving many custom operations and modules. Compared with pure DL methods, this might reduce hardware optimization and parallel computing efficiency, leading to relatively longer inference times. Nevertheless, our method achieves the best performance and demonstrates superior generalization across three datasets. Overall, the tradeoff between performance and computational complexity is within an acceptable range.

E. Discussion

From the above extensive experimental results, it is evident that our proposed method has significant advantages in several aspects. First, the architecture of the MSCSCNet is highly interpretable, with transparent and traceable internal mechanisms and information flow. Each operation is model-driven, ensuring that the extracted coefficient features have intuitive physical meanings, facilitating targeted future improvements. Moreover, by explicitly distinguishing between PSCs and SSC, the multiscale feature extraction process becomes more interpretable, reducing redundancy while obtaining more compact and goal-oriented coefficient features. This improves detection in large continuous change areas and ambiguous boundaries. Furthermore, the introduction of CS-TSSD allows MSCSCNet to learn from large amounts of unlabeled data, enhancing its sensitivity and robustness in CD.

In addition, by combining model-based and DL approaches, our method excels on individual datasets and generalizes better across different datasets, even with limited labeled data. As shown in Tables II–IV, our method achieves the best performance across all datasets, while other pure DL methods show unstable performance with fluctuating rankings. This instability is due to their reliance on empirical and subjective designs, leading to significant architectural differences and varying adaptability to different datasets. Moreover, these methods are prone to overfitting with limited data, resulting in inconsistent

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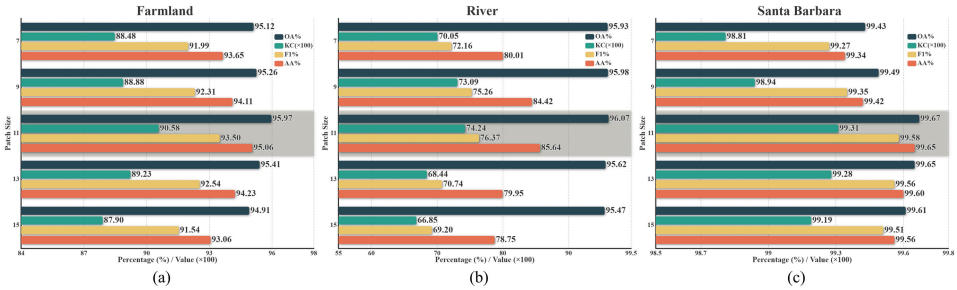


Fig. 11. Experimental results of different input patch sizes on three datasets. (a) Farmland dataset. (b) River dataset. (c) Santa Barbara dataset.

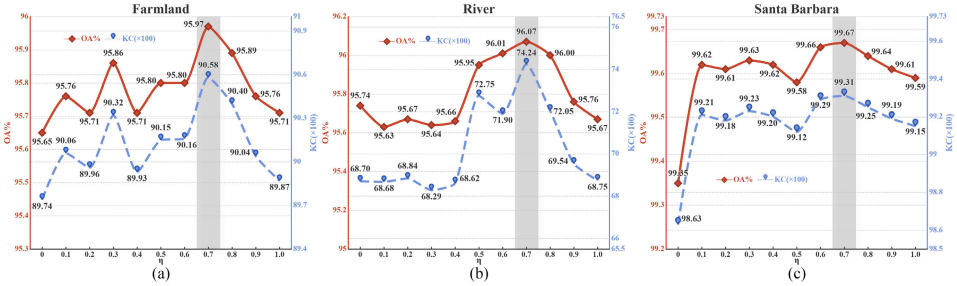


Fig. 12. Experimental results of different tradeoff parameters η applied to CS-TSSD on three datasets. (a) Farmland dataset. (b) River dataset. (c) Santa Barbara dataset.

TABLE V

QUANTITATIVE ASSESSMENT OF DIFFERENT FEATURE FUSION METHODS ON THE THREE DATASETS

Dataset	Method	OA%	KC(×100)	F1%	AA%
Farmland	Add	95.15	88.62	92.12	93.92
	Sub	**95.97**	**90.58**	**93.50**	**95.06**
	Concat	95.48	89.47	92.75	94.64
River	Add	94.77	60.92	63.60	75.67
	Sub	**96.07**	**74.24**	**76.37**	**85.64**
	Concat	95.67	69.83	72.14	81.55
Santa Barbara	Add	99.15	98.21	98.92	99.10
	Sub	**99.67**	**99.31**	**99.58**	**99.65**
	Concat	99.42	98.78	99.26	99.41

TABLE VI

QUANTITATIVE ASSESSMENT OF DIFFERENT SCALES OF SSC ON THE THREE DATASETS

Dataset	Scale	OA%	KC(×100)	F1%	AA%
Farmland	S	95.42	89.25	92.55	94.24
	M	**95.97**	**90.58**	**93.50**	**95.06**
	L	95.20	88.72	92.17	93.87
River	S	95.83	69.54	71.71	79.99
	M	**96.07**	**74.24**	**76.37**	**85.64**
	L	95.50	65.82	68.10	77.30
Santa Barbara	S	99.58	99.11	99.46	99.51
	M	**99.67**	**99.31**	**99.58**	**99.65**
	L	99.58	99.12	99.46	99.53

TABLE VII

QUANTITATIVE ASSESSMENT OF THE EFFECT OF THE PROPOSED PSCUM AND SSCUM ON THE THREE DATASETS

Dataset	PSCUM	✓	✗	✓
	SSCUM	✗	✓	✓
Farmland	OA%	95.68	95.56	**95.97**
	KC(×100)	89.82	89.60	**90.58**
	F1%	92.94	92.82	**93.50**
	AA%	94.42	94.53	**95.06**
River	OA%	95.71	95.41	**96.07**
	KC(×100)	68.46	65.58	**74.24**
	F1%	70.69	67.93	**76.37**
	AA%	79.33	77.57	**85.64**
Santa Barbara	OA%	99.54	99.46	**99.67**
	KC(×100)	99.03	98.88	**99.31**
	F1%	99.41	99.32	**99.58**
	AA%	99.46	99.39	**99.65**

TABLE VIII

QUANTITATIVE ASSESSMENT OF DIFFERENT PENALTY FACTORS ON THE THREE DATASETS

Dataset	γ	OA%	KC(×100)	F1%	AA%
Farmland	0	95.57	89.56	92.74	94.20
	0.4	95.72	89.94	93.03	94.52
	0.8	95.64	89.75	92.89	94.37
	1.2	95.95	90.50	93.43	94.94
	1.6	**95.97**	**90.58**	**93.50**	**95.06**
	2	95.80	90.14	93.18	94.71
River	0	95.69	70.64	72.97	82.70
	0.4	95.62	67.85	70.13	79.14
	0.8	95.80	71.54	73.81	83.23
	1.2	95.96	70.44	72.54	80.34
	1.6	**96.07**	**74.24**	**76.37**	**85.64**
	2	95.91	71.71	73.91	82.70
Santa Barbara	0	99.51	98.98	99.38	99.46
	0.4	99.54	99.03	99.41	99.52
	0.8	99.57	99.10	99.45	99.54
	1.2	99.65	99.26	99.55	99.62
	1.6	**99.67**	**99.31**	**99.58**	**99.65**
	2	99.60	99.17	99.49	99.58

performance across datasets. In contrast, our method leverages both model-based and DL advantages, incorporating prior information and structured constraints to build models that capture more fundamental features.

However, it is noteworthy that, despite moderate Params and FLOPs, the inference time remains relatively long. Therefore, future work will concentrate on model compression and acceleration optimization techniques to achieve a more hardware-friendly and efficient architecture. Moreover, the current method is primarily applied to homogeneous HSI-CD, and extending it to CD between heterogeneous bitemporal images is also an important direction for future research.

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TABLE IX
COMPUTATIONAL COSTS ON THE FARMLAND DATASET

Method	Params (M)	FLOPs (G)	Inference (s)
D-LIEG	31.44	430.78	4.58
D^2AGCN	2.15	3.30	7.91
MSDFFN	26.46	2.32	3.72
DA-Former	0.40	0.04	2.93
MFCEN	153.82	0.84	7.94
Ours	16.55	9.35	13.50

IV. CONCLUSION

In this article, we have explored the potential of integrating DL and model-based approaches for HSI-CD, culminating in a semi-supervised MSCSCNet. Specifically, we introduce an MSCSC model to capture both SSC and PSCs at various scales, reducing information redundancy while extracting multiscale features. We employ the ADMM to solve the optimization problem of this model, deriving the iterative update algorithms. This algorithm is then unfolded to establish a deep interpretable network, MSCSCNet, as the encoder, which combines clear physical interpretability with deep feature extraction. Subsequently, we adopt a two-stage semi-supervised training strategy to train the MSCSCNet, utilizing a hybrid training mechanism of SL and CS-TSSD, where the latter enables the network to learn from a vast amount of unlabeled data and remain sensitive to change components. Extensive experiments on three benchmark datasets have demonstrated the effectiveness of our proposed method, achieving state-of-the-art performance compared with the existing DL algorithms.

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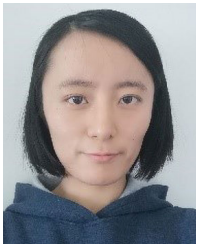
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